

Aamir Syed

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Engineering, Minor in Math

August 2024 - December 2027

- **Coursework:** Data Structures and Algorithms, Advanced Algorithms, Discrete Math, Objects and Design, Intro to Embedded Systems, Malware Reverse Engineering, GPU Programming, Computer Security, Object-Oriented Programming, Advanced Malware Analysis

EXPERIENCE

Software Engineering Intern

June 2025 – April 2026

MARGA Lab

Atlanta, GA

- Deployed and managed **AWS EC2 instances** to host data-processing clusters and to acts as a remote server for Wireshark Analysis.
- Implemented a GitHub Actions **CI/CD pipeline** to automate Docker image builds, ECR pushes, and Kubernetes deployments on every commit to main.
- Engineered a high-precision system with LEO satellite data using **Python** and the SGP4 algorithm to propagate TLE data, achieving a **64% accuracy improvement** over previous iterations.

AI Research Assistant

August 2024 – October 2024

University of Washington, Seattle

Seattle, WA

- Utilized **Tensorflow** to measure computational efficiencies of **Spiking Neural Networks (SNNs)** and **Neuromorphic Computing**.
- Evaluated SNN performance specifically for **Time Series Forecasting**, with **temporal dynamics** to improve prediction accuracy by **7%**.

Software Engineering Fellow

June 2024 – August 2024

Jane Street

New York City, New York

- Studied **combinatorics**, **number theory**, and **algorithmic programming** to solve complex computational problems.
- Developed an **automated trading bot** for the electronic trading challenge, achieving a performance yield that **outperformed 75% of competitors**.
- Engineered a **Wordle solver** using Python with **Shannon entropy** to solve for expected information gain, optimizing the opening guess and reducing the average solve rate to ≈ 3.4 guesses.
- Built a **J2EE** web application using **Java Servlets** and **JSP** with **JDBC** for MySQL persistence, implementing MVC separation and session-based authentication

PROJECTS

Deep RL Chess Engine | *Pytorch, FastAPI, React*

January 2026 – February 2026

- Trained a chess engine by implementing policy network in PyTorch, enabling self-improving gameplay through self-play reinforcement learning
- Built a full-stack chess application by integrating a React UI with a FastAPI backend and deploying GPU training on Georgia Tech's PACE-ICE HPC cluster

ESP32 Spotify Display | *Embedded, C++, AWS*

March 2025 – April 2025

- Created an embedded project that shows what I'm listening to on Spotify through an AWS Lambda function
- Uses the SPI peripheral to communicate with and FreeRTOS's tasks for non-blocking updates to the screen

FitJacket | *React.js, Node.js, Express.js, SQL, Google Gemini API, Tailwind CSS, AWS*

May 2025 – June 2025

- Worked with a team to create an intelligent meal recommendation engine by integrating **Google Gemini 1.5 Pro** via the **Google Generative AI SDK**, to generate 7-day nutritional plans based on user-specific macros and fitness goals.
- Built and documented a **RESTful API** to bridge the frontend with a **SQL database**, ensuring data persistence and optimized query performance.
- Developed an interactive dashboard using **React.js** and **Tailwind CSS**, featuring data visualizations and progress charts to enhance user retention and engagement.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, C, SQL (Postgres), JavaScript, HTML/CSS, VHDL, Verilog, RISC-V, x86

Frameworks: React, Node.js, Express.js, .NET, LangChain

Developer Tools: Git, Docker, IDA Pro, Wireshark, openssl, Bash, Linux, Intel Quartus, Ghidra, MongoDB, AWS, Prometheus

Libraries: Pytorch, Tensorflow